INCH-POUND
MIL-DTL-24643/60A
28 January 2010
SUPERSEDING
MIL-DTL-24643/60
9 September 2005

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LSC5 AND LSC5W

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

This cable can be manufactured in two variations: LSC5 (Unshielded (UTP), Non-Watertight) and LSC5W (Unshielded (UTP), Watertight).

Construction (Non-Watertight and Watertight)

First	-	Copper conductor, solid, 24 AWG nominal.
Second	-	Polyolefin insulation. Wall thickness as necessary, in order for the completed cable to comply with the specified electrical requirements. Maximum insulation diameter shall not exceed 0.046 inch.
Third	-	Two insulated conductors, twisted together such that they comply with the electrical requirements specified herein. Wire identification codes shall be in compliance with ANSI/TIA-568-C.2 as follows: Pair 1 – white/blue and blue Pair 2 – white/orange and orange Pair 3 – white/green and green Pair 4 – white/brown and brown

Construction Variant for Type LSC5

Fourth	-	Four pairs cabled together with a lay as required to comply with electrical requirements.
Fifth	-	Cross-linked polyolefin jacket. Minimum thickness of 0.040 inch.

Construction Variant for Type LSC5W

Fourth	-	Four pairs cabled together with a lay as required to comply with electrical requirements.
Fifth	-	For watertight variant, filler material shall comply with the requirements of 3.3.5 and 3.3.5.3 of MIL-DTL-24643.
Sixth	-	An optional binder.
Seventh	-	Cross-linked polyolefin jacket. Minimum thickness of 0.040 inch.

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TABLE I. Details.

Military part no. M24643/60	Type	Number of pairs	Overall diameter (in.)	
			Min.	Max.
-01UN	LSC5-4	4	0.220	0.270
-02UN	LSC5W-4	4	0.230	0.290

REQUIREMENTS:

Qualification Required.

INSPECTION:

Basic Electricals:

DC resistance (ohms/100 m, max.)	9.38
DC resistance unbalance (max.)	5%
Jacket flaws	No failure
Conductor continuity	No failure
Voltage withstand	
Conductor to conductor	1.5kVAC or 2.5kVDC

Group A:

Visual and dimensional	No failure
Watertightness (Type LSC5W-4 only)	No failure
Capacitance	
Mutual capacitance of a pair (nominal @1 kHz)	5.6 nf/100 m
Pair to ground capacitance unbalance (max. @1 kHz)	330 pf /100 m
Input impedance (1 MHz to 100 MHz)	100 ohms ±15%

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TABLE II. Swept electrical parameters.

Frequency (MHz)	**1.0**	**4.0**	**10**	**16**	**31.25**	**62.5**	**100**
Return loss dB/100 m (min.)	20.0	23.0	25.0	25.0	23.6	21.5	20.1
Insertion loss dB/100 m (max.)	2.0	4.1	6.5	8.2	11.7	17.0	22.0
NEXT dB/100 m (min.)	65.3	56.3	50.3	47.2	42.9	38.4	35.3
ELFEXT dB/100 m (min.)	63.8	51.8	43.8	39.7	33.9	27.9	23.8
PS NEXT dB/100 m (min.)	62.3	53.3	47.3	44.2	39.9	35.4	32.3
PS ELFEXT dB/100 m (min.)	60.8	48.8	40.8	36.7	30.9	24.9	20.8
Propagation delay ns/100 m (max.)	570	552	545	543	540	539	538
Delay skew ns/100 m (max.)	45	45	45	45	45	45	45
NOTE: Values are for reference only. Actual values shall be determined utilizing the formulas in ANSI/TIA-568-C.2.							

Group B:

Drip (95±1 °C) Zero

Tear (lb/in thickness, min.) (ASTM D470) 35

Physicals (unaged)

Insulation

Tensile strength (lb/in^2, min.) 700

Elongation (percent, min.) 160

Jacket

Tensile strength (lb/in^2, min.) 1300

Elongation (percent, min.) 160

Thermoset proof test (percent, max.)

Jacket 50

Group C:

Physicals (aged air oven)

Insulation (polyolefin)

Tensile strength (percent of unaged, min.) 75

Elongation (percent of unaged, min.) 75

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Jacket	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Shrinkage	No failure
Cable filler removability	No failure
Heat distortion (percent of unaged, max.)	30
Conformance to material, construction and coverage	No failure
Permanence of printing (jacket) (cycles, min.)	125
Group D:	
Flame propagation (cable)	No failure
Qualification Inspection:	
Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:	
Aging and compatibility (cable) (95±1 °C)	No failure
Abrasion resistance (jacket)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	8
Insulation	18
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index, max.	
Jacket	25
Fillers	55
Insulation	45
Toxicity index, max.	
Jacket	5
Fillers	5
Insulation	1.5

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Durometer (jacket) – Type A (hardness, min.)	80
Weathering (jacket)	No failure
Temperature humidity (LSC5W only)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSC5, LSC5W	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army – MI
Navy – SH

Preparing Activity:
Navy – SH
(Project 6145-2009-103)

Review Activities:
Army – AR, AV, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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